ANALYZING PITCHER DECEPTION AND COMMAND

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OUR TEAM

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AGENDA

1. Business Background
2. Quantifying Deception & Command
3. Graph Neural Network (GNN)
4. Conclusions

EVOLUTION OF BASEBALL DECISION MAKING

"Traditional" Era	Moneyball	Largescale Analytics and ML
Batting Avg., Strikeouts →	On Base %, Slugging →	WAR, Spin Rate

"Traditional" Era

- Decision making based on **watching, physical tools,** and **results-based metrics**
- Scouts and GMs had to be long-standing baseball players or coaches

Moneyball

- Teams utilize **more complex stats** and dedicated data analysts to find undervalued players

Largescale Analytics and ML

- **Large scale biomechanical data** now available
- More data = more advanced metrics = **even more analysis and modeling** used to acquire and train players

WHICH PITCHER IS BETTER?

Pitcher A

Fastball: 96 MPH

Curveball: 82 MPH with 13in of downward break

Cutter: 91 MPH

Changeup: 86 MPH

4.2 Earned Runs Allowed per 9 Innings (ERA)

Pitcher B

Fastball: 92 MPH

Curveball:75 MPH with 10in of downward break

Sinker: 90 MPH

Changeup: 80 MPH

3.2 Earned Runs Allowed per 9 Innings (ERA)

How do we explain this difference?

DECEPTION AND COMMAND – THE "SECRET STUFF"

- **Deception** is a pitcher's ability to make a pitch's speed and location **difficult to judge or misleading for the batter**.

- **Command** is a pitcher's ability to make a pitch's speed and location **correspond with their desired speed and location.**

- Better deception and command = pitches harder to hit well = fewer base hits/walks = more games won = more revenue.

BUSINESS PROBLEM

- **Context:** In 2025, the White Sox had the 3rd lowest MLB team salary. As such, they benefit particularly from finding undervalued pitchers who outperform traditional metrics through **command and deception.**
- **Problem:** Though complex biomechanical and result data is available for pitches, **command and deception lack any sort of standard quantification** despite being referenced frequently.
- **Plan:** By exploring and modeling the data, we plan to create **robust metrics for command and deception**, along with an **analysis of their indicators.**
- **Opportunity:** The insights from this project will aid both **player development and acquisition.**

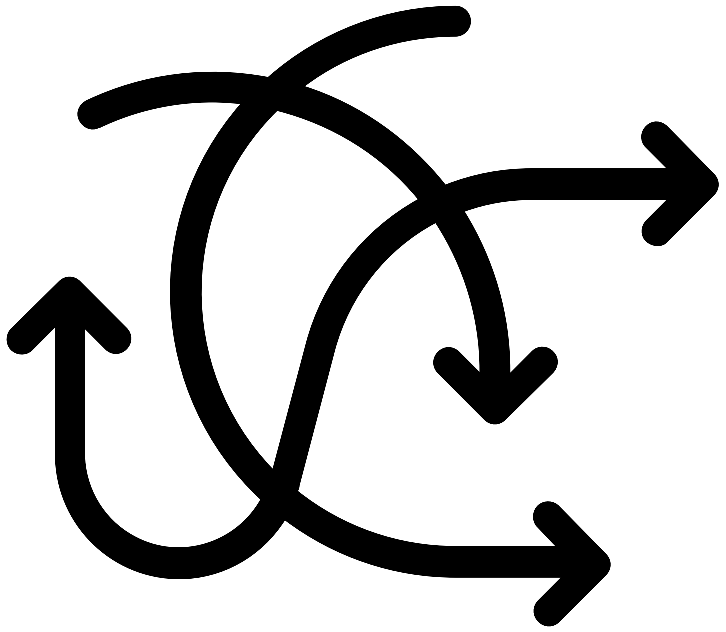
DATA ANALYSIS

Deception Quantification

Command Quantification

GNN Model on Biomechanical Data

DECEPTION QUANTIFICATION

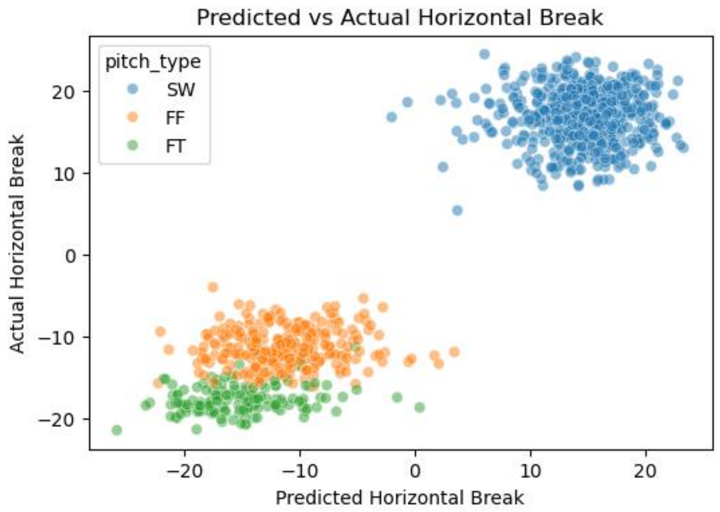


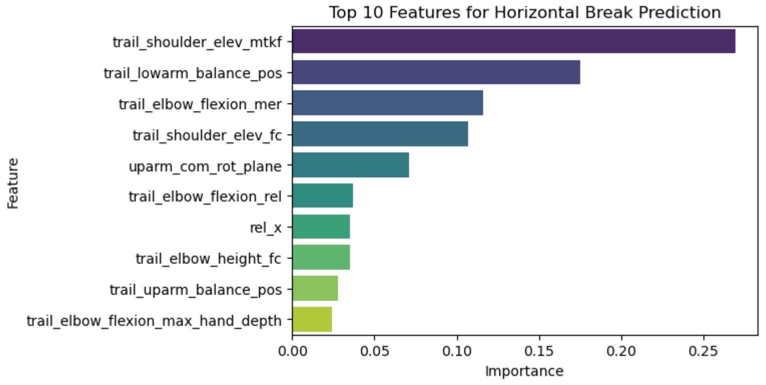
Key Questions:

1. How does a pitch's velocity and break compare to what the batter expects?
2. At what angle is the ball entering the strike zone?
3. How well does a pitcher do at taking away time from the batter to react?

HOW MUCH DOES A PITCH BREAK VS EXPECTED?

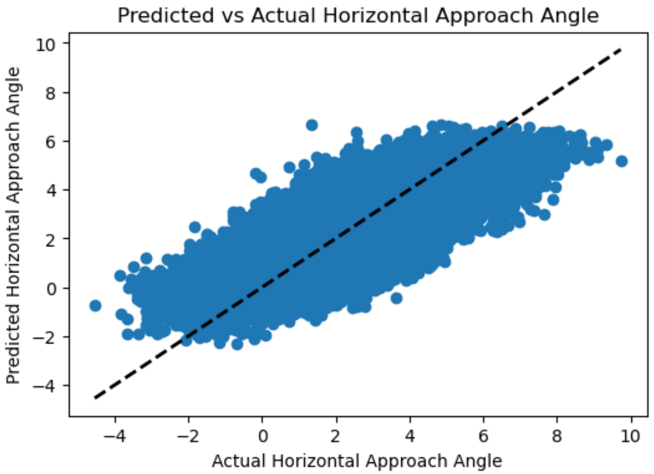
- **Methodology:** pitcher-specific XGBoost models that predict horizontal and vertical break. Each model weighted by pitch type frequency to avoid accidentally capturing pitch frequency and selection
- **Why?** A pitcher aims to hide information from the batter prior to release. **If our model can identify key patterns this may indicate that a batter can visually pick up on key characteristics**
- **Results:**
  - Accuracy varies by pitcher (expected and what we want) Struggles to make accurate predictions for players with large break distances
  - **RMSE** for each pitcher gives us an indicator for deception





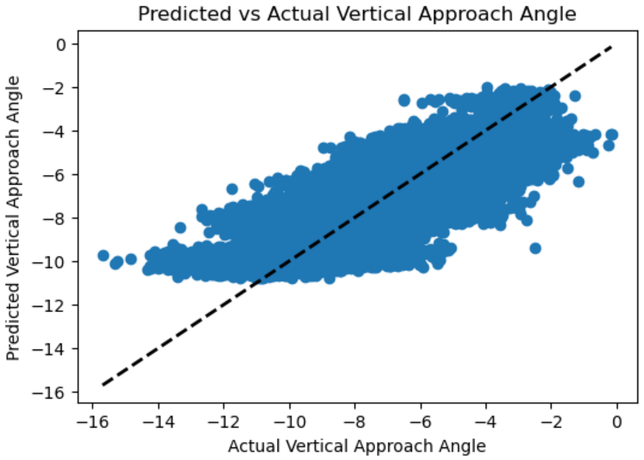
HORIZONTAL AND VERTICAL APPROACH ANGLE

- **Goal:** predict the horizontal and vertical approach angle of a pitch given the pitcher's height, arm slot, release angle
- **Why?:** approach angles represent the angle at which the ball comes into the strike zone. If it differs a lot from expected that may mean a pitch breaks late, which deceives the batter
- **Results:** Depending on pitch type each of **HAA and VAA over expected are significant predictors of deceptive outcome metrics**
  - For example, on fastballs VAA over expected is significant but for sliders VAA over expected is not.



R^2: .677

R^2: .691

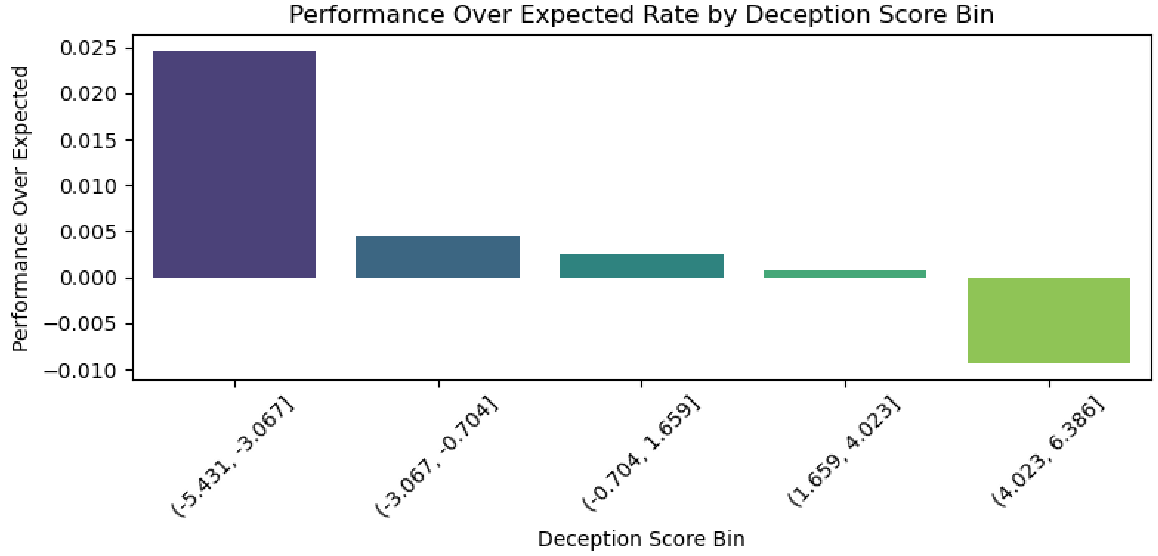


ADDING IN OTHER FACTORS AND SCORING PITCHES

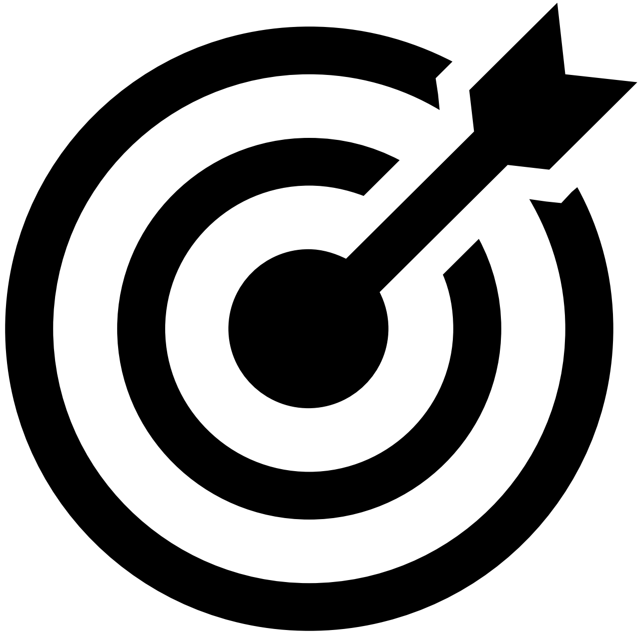
Methodology:

- In addition to the break and approach angle differences we add **time with ball hidden** and **extension**
- Use these metrics to predict outcome metrics like called strike + whiff rate (CSW), performance over expected, and swing differences
- Use t-values to weight each metric and create a final deception score

Results:



COMMAND QUANTIFICATION



Key Question:

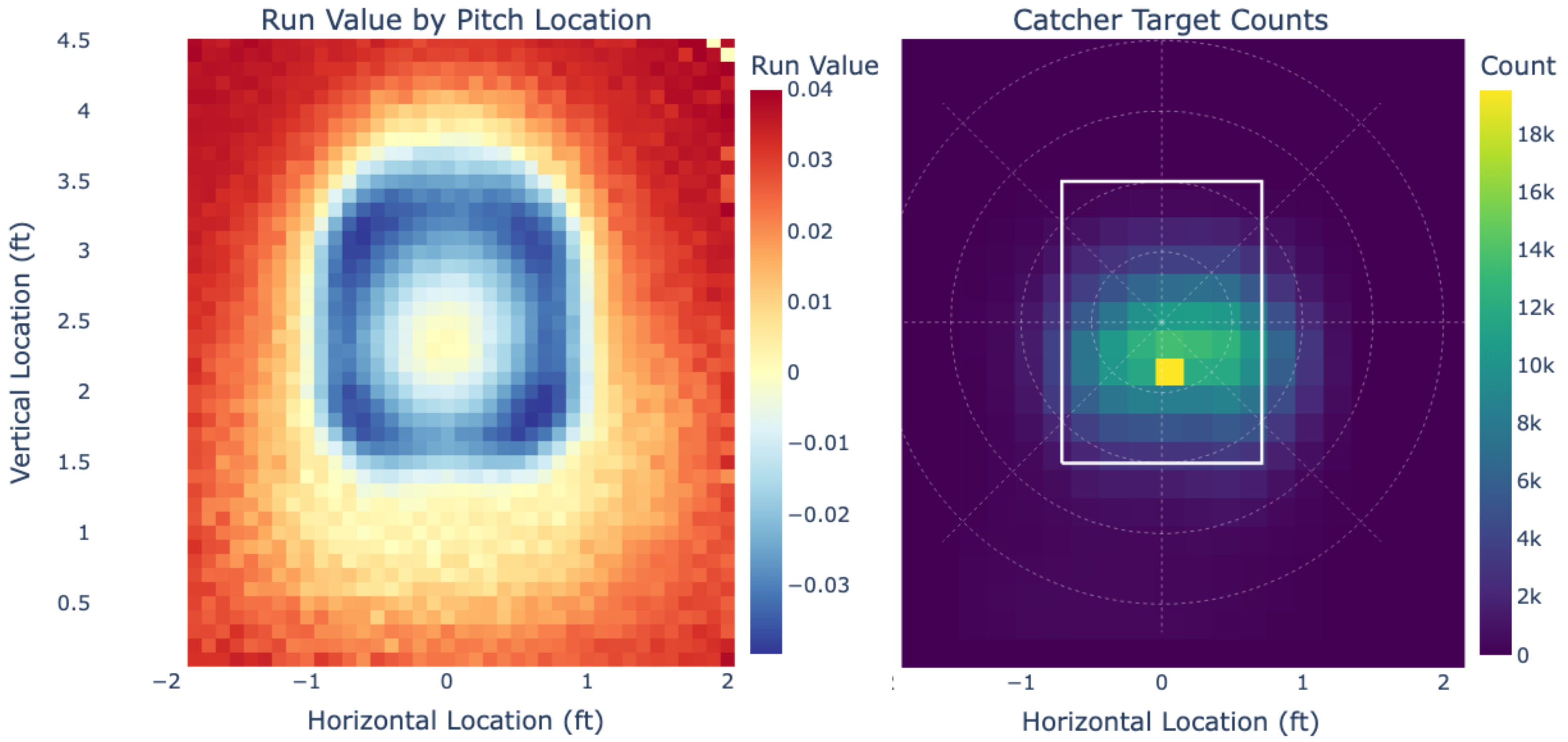
How can we infer intended pitch location?

Two Approaches:

1. Utilize catcher target as a proxy.
2. Infer directly from location patterns.

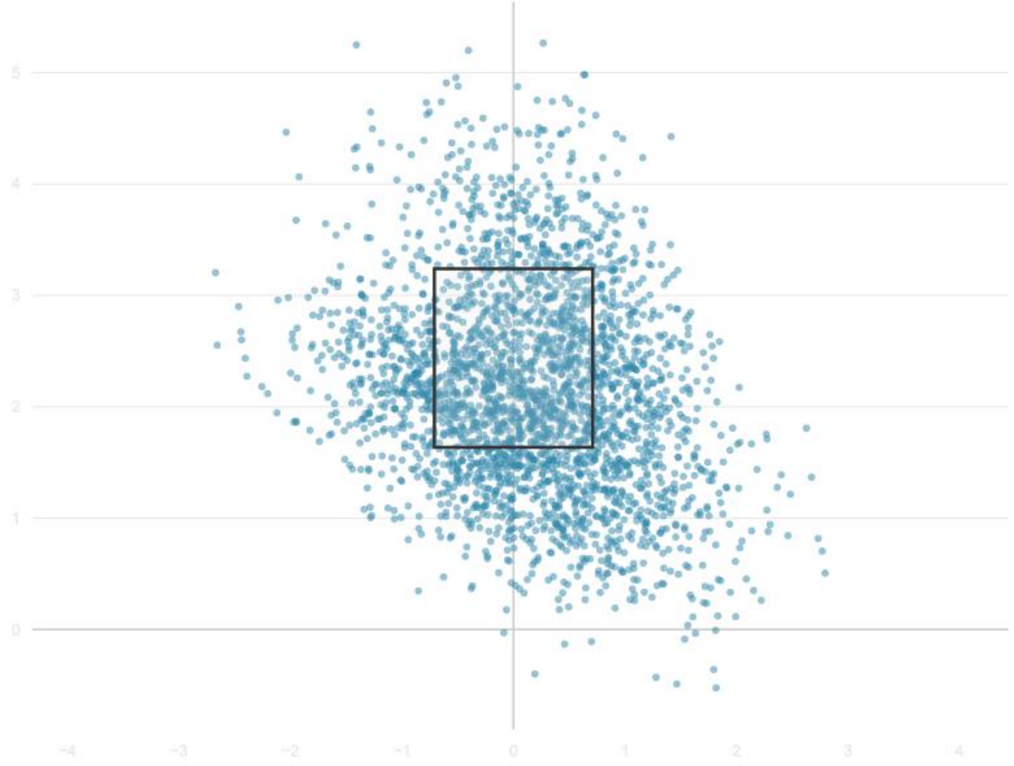
CATCHER TARGET & POLAR COORDINATES

- **Conversion to polar coordinate system**
- Key New Features:
    - Radial Difference
    - Angular Offset
    - **Highly predictive of run value + damage**
- Catcher Target Limitations:
    - Low Variance
    - Tracking Errors
    - Weak Assumptions

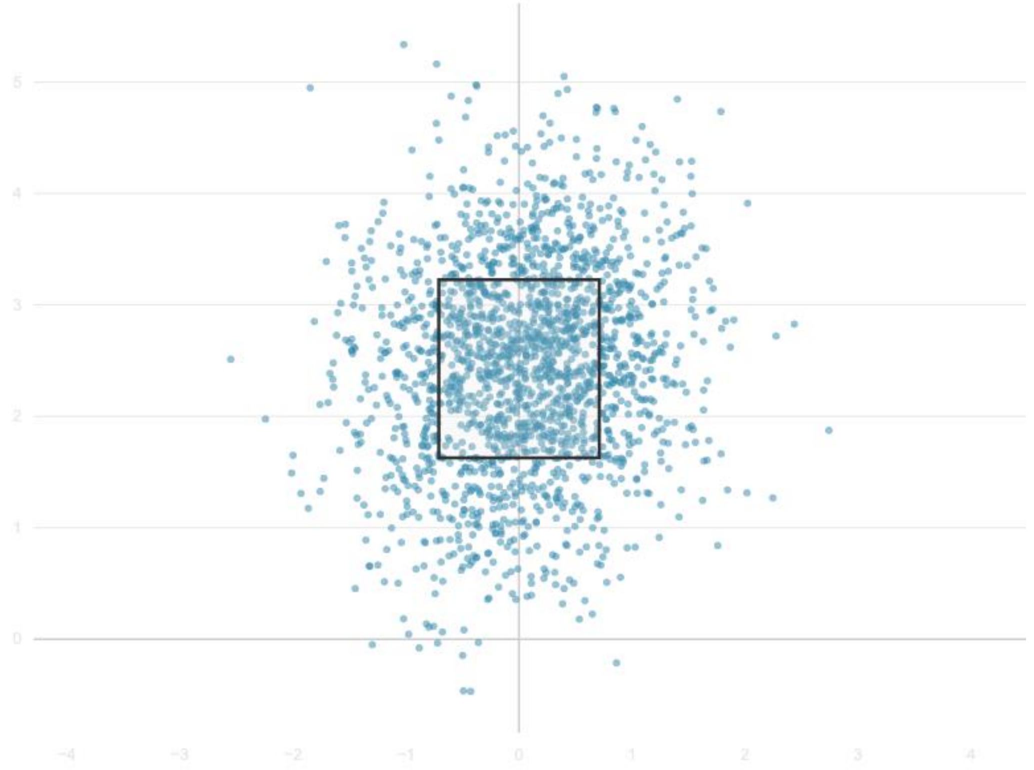


PITCH SPREAD – NEAR RANDOM PROCESS

PITCHER 1: ?

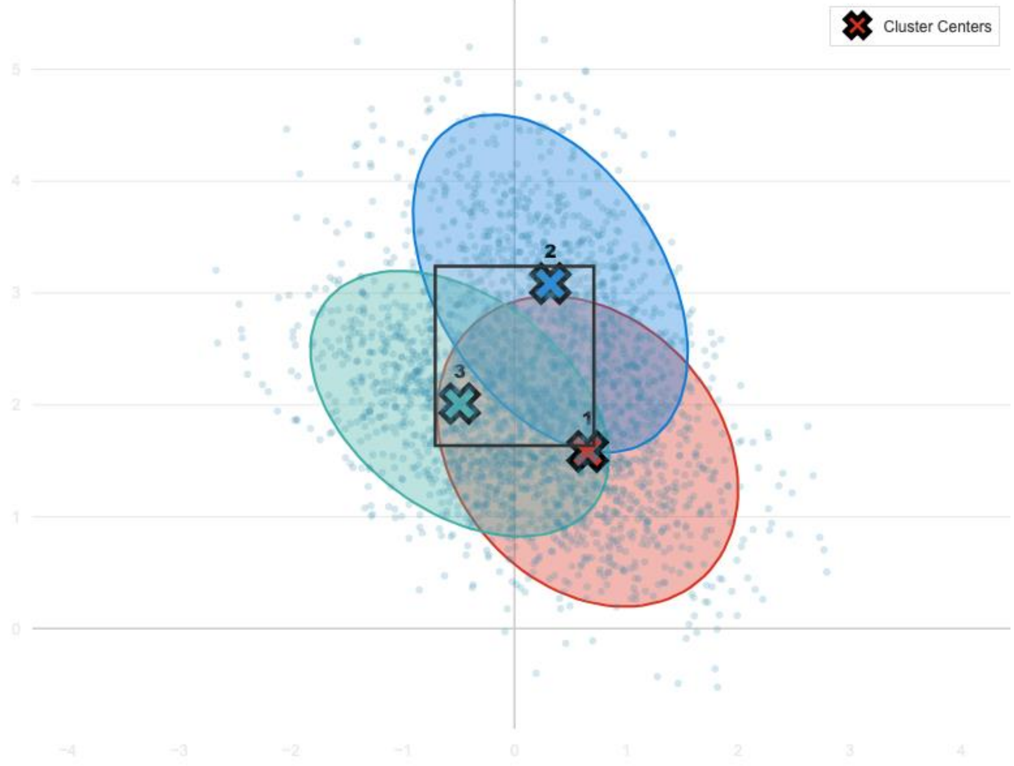


PITCHER 2: ?

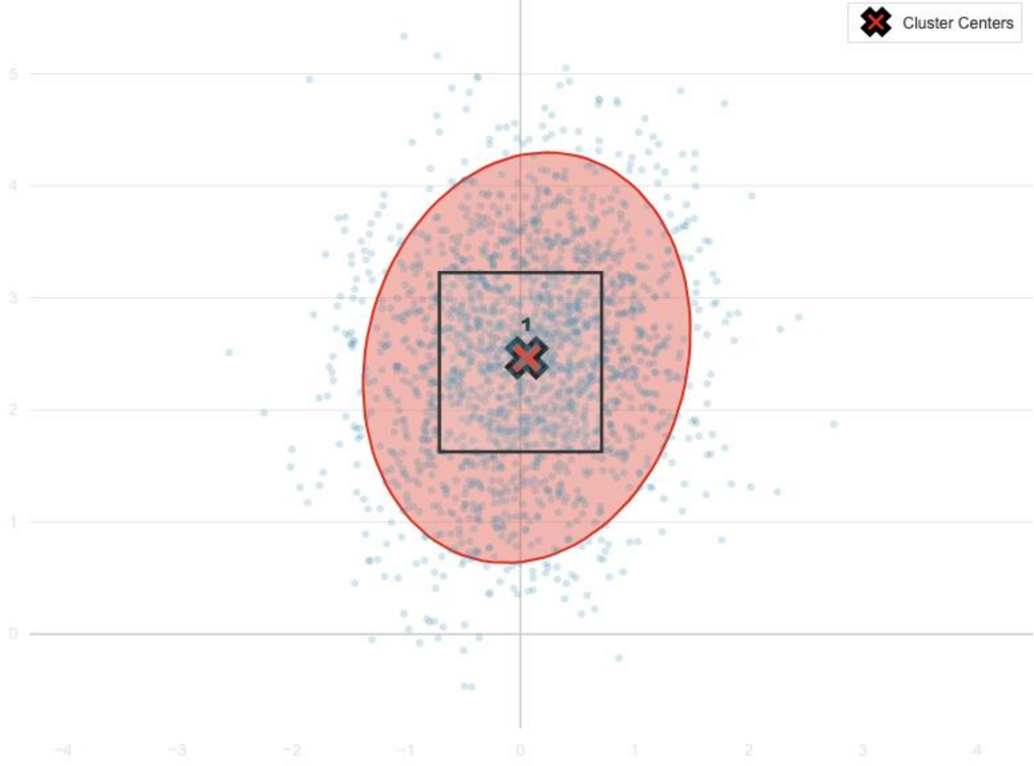


PITCH SPREAD – GAUSSIAN MIXTURE MODELING

PITCHER 1: AARON NOLA (RANK – 5)



PITCHER 2: TARIK SKUBAL (RANK – 390)



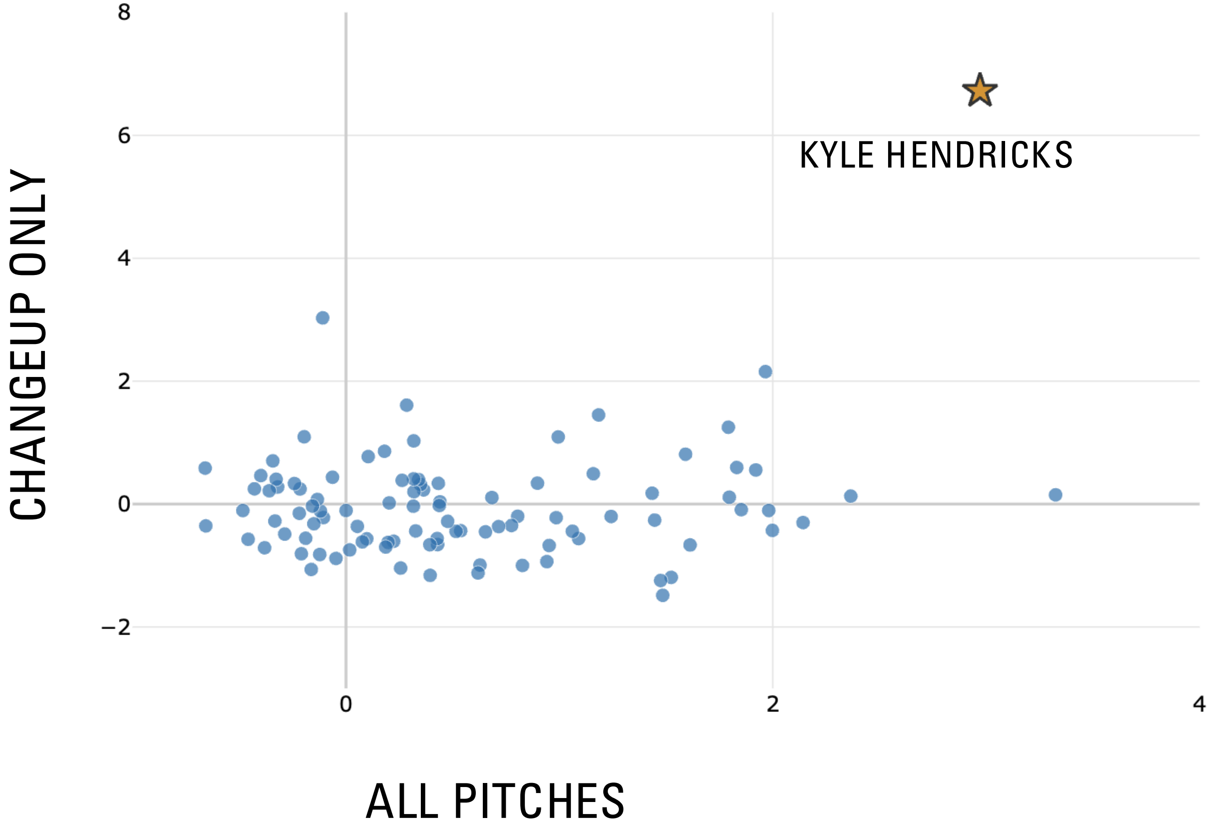
CASE STUDY – KYLE HENDRICKS' CHANGEUP

"I think you'll wait a long time before you get the next Kyle Hendricks. His command was exceptional. His changeup was exceptional."

- *Jed Hoyer – (President of Baseball Operations for the Chicago Cubs)*

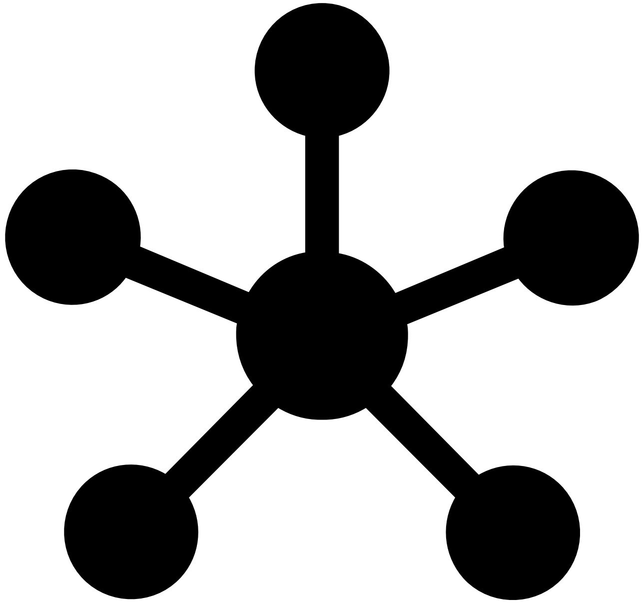
CASE STUDY – KYLE HENDRICKS' CHANGEUP

COMMAND SCORES - FULL ARSENAL VS. CHANGEUP



*Among pitchers that threw 200+ changeups in 2024

GNN MODEL ON BIOMECHANICAL DATA



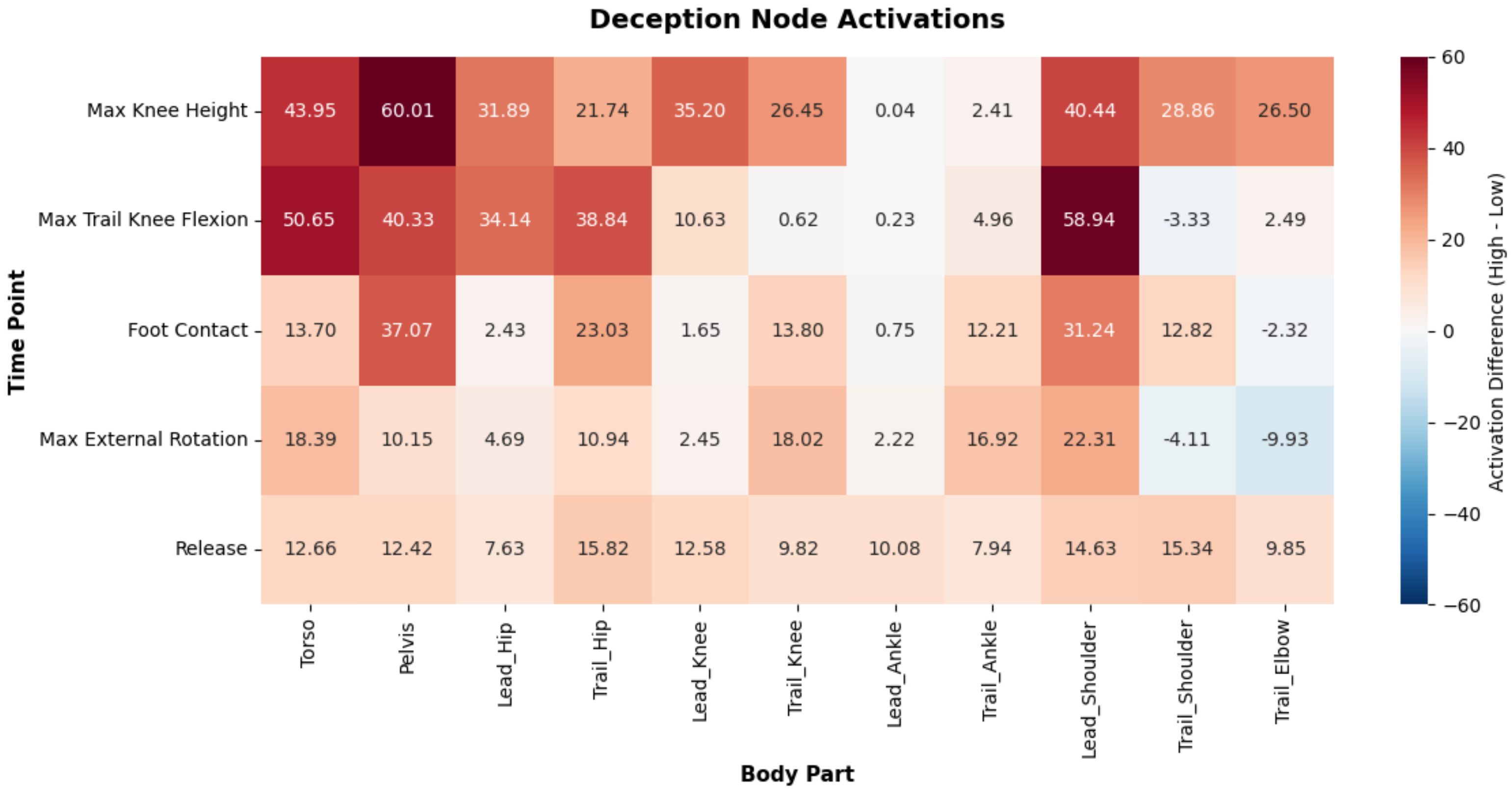
GRAPH NEURAL NETWORKS

- Once we have scores, a useful next step is to look at what **biomechanical features** contribute to high or low command/deception, and at which points during the pitch delivery
- Graph Neural Networks enable us to answer this by **modeling a pitcher's body as a graph**
- Despite being more complex than more traditional tree-based methods, we believe that GNNs have the highest potential due to their unique fit to our business problem

GNN RESULTS AND TAKEAWAYS

Deception

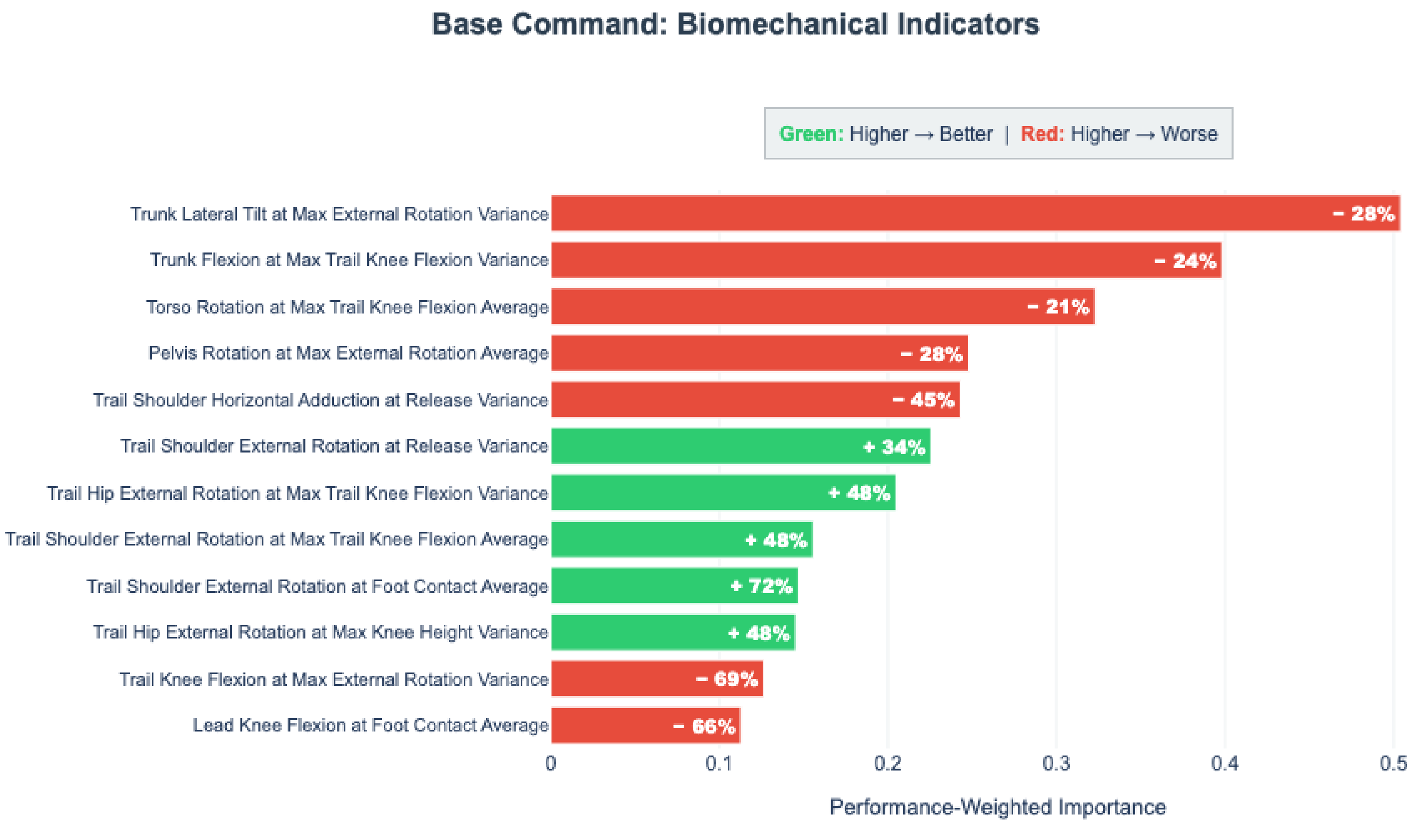
- **Moderate signal detected!**
- R-squared: 0.17
  - Good for noisy data
- Analyzed node activations for highly deceptive pitches.
- Early upper body posture important.



GNN RESULTS AND TAKEAWAYS

Command

- Data too noisy to yield signal at pitch level.
- Pivot to random forest classifiers at the pitcher level.
- Engineered Features for variance in pitcher delivery.
- Balance and Consistency are key.



WHICH PITCHER IS BETTER?

Patrick Corbin
Pitcher A

Fastball: 96 MPH

Curveball: 82 MPH with 13in of downward break

Cutter: 91 MPH

Changeup: 86 MPH

Command Score: 59th
Deception Score: 290th

4.2 Earned Runs Allowed per 9 Innings (ERA)

Ranger Suarez
Pitcher B

Fastball: 92 MPH

Curveball:75 MPH with 10in of downward break

Sinker: 90 MPH

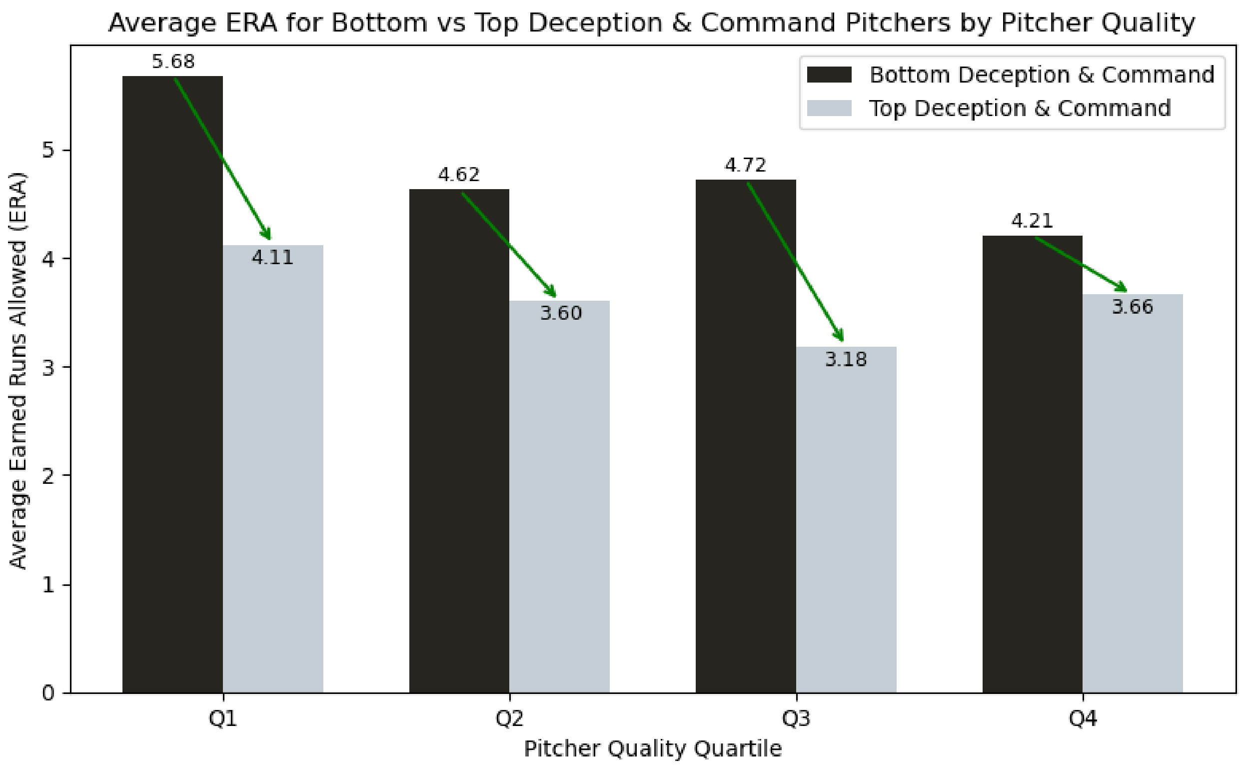
Changeup: 80 MPH

Command Score: 18th
Deception Score: 38th

3.2 Earned Runs Allowed per 9 Innings (ERA)

TREA TURNER ON RANGER SUÁREZ

“'There are some guys who throw 100 and you see the ball good and there's some guys who throw 90 and you don't see the ball well,' Turner said. 'When you're watching on T.V. it doesn't make sense, but he's one of those guys where everything looks the same out of his hand and it takes longer to pick things up. By the time you make your decision, it's a little too late. So, it's not always velo. **It's more control, deception and pitch mix and that's where I think he can separate himself.'**”



For pitchers of similar quality, having better deception and command reduces runs allowed by **~0.7 per game**. This results in **113 fewer runs** leading to **11 more wins for the White Sox.**

THANK YOU – LET'S GO SOX!